

Topic B17

# Variation and selection

**Concept guide · Worked application · Cambridge-style practice · Fully explained answers**

## What this guide teaches

A complete explanation of the syllabus concepts, biological mechanisms, common misconceptions, data interpretation and original exam-style practice for this topic.

## Syllabus map

|       |                    |
|-------|--------------------|
| B18.1 | Variation          |
| B18.2 | Mutation           |
| B18.3 | Adaptive features  |
| B18.4 | Natural selection  |
| B18.5 | Selective breeding |

How to study: retrieve definitions, cover the application, then complete the practice paper without using the notes.

# 1. Variation

Continuous variation shows a range and is often controlled by many genes plus environment, such as height. Discontinuous variation has distinct categories and is often controlled by one or few genes, such as ABO blood group. Genetic variation arises from mutation, meiosis and fertilisation.

## Examiner insight

Choose an appropriate graph: histogram for continuous data, bar chart for categories.

# 2. Mutation

Mutation is a random change in genetic material. Gene mutations alter DNA base sequence; chromosome mutations change structure or number. Mutations can be harmful, neutral or beneficial depending on environment. Ionising radiation and some chemicals increase mutation rate.

## Examiner insight

Mutations arise randomly; the environment selects among resulting phenotypes.

# 3. Adaptation and selection

Adaptive features improve survival and reproduction in a particular environment. Natural selection requires variation, selection pressure, differential survival and reproduction, inheritance and change in allele frequency over generations. Evolution is population change, not purposeful change by individuals.

## Examiner insight

Use “individuals are selected; populations evolve”.

# 4. Selective breeding

Humans choose parents with desired phenotypes, breed them and select offspring over many generations. It can increase yield or disease resistance but reduce genetic variation and increase inherited defects. Artificial selection and natural selection differ mainly in the selecting agent.

## Examiner insight

Evaluate both agricultural benefit and genetic risk.

## Worked application

Situation: A drought persists for many generations.

Prediction: Alleles for water-conserving features may increase.

Reason: Individuals with advantageous inherited variation survive and reproduce more.

# Exam technique

## High-value habits

Define terms precisely, describe visible patterns before explaining them, link structure to function, use data values as evidence, identify variables explicitly and answer the command word directly.

## Retrieval checklist

- Recall five key definitions without notes.
- Explain one process as a linked cause-and-effect chain.
- Interpret the data set and identify a limitation.
- Complete the timed practice and mark only after finishing.

# Practice Paper

Topic B17

## Variation and selection

### Instructions

Answer all questions. Select one option for multiple choice. Use values from tables where relevant. Write explanations on the lines. Individual parts carry 1-3 marks.

Candidate name:

Class / date:

**Total marks: 50**

Suggested time: 60 minutes.

## Section A - Multiple choice

Choose the best answer for each question.

1. Which statement best defines variation? [1]
  - A differences between individuals of the same species
  - B a random change in genetic material
  - C an inherited feature increasing fitness in an environment
  - D differential survival and reproduction caused by environmental pressures
2. Which statement best defines mutation? [1]
  - A an inherited feature increasing fitness in an environment
  - B a random change in genetic material
  - C differential survival and reproduction caused by environmental pressures
  - D human selection of parents with desired features
3. Which statement best defines adaptation? [1]
  - A differential survival and reproduction caused by environmental pressures
  - B human selection of parents with desired features
  - C an inherited feature increasing fitness in an environment
  - D differences between individuals of the same species
4. Which statement best defines natural selection? [1]
  - A human selection of parents with desired features
  - B differences between individuals of the same species
  - C a random change in genetic material
  - D differential survival and reproduction caused by environmental pressures
5. Which statement best defines selective breeding? [1]
  - A human selection of parents with desired features
  - B differences between individuals of the same species
  - C a random change in genetic material
  - D an inherited feature increasing fitness in an environment
6. Which statement about variation is correct? [1]
  - A Differences only between species.
  - B Continuous variation shows a range and is often controlled by many genes plus environment, such as height.
  - C A directed change caused by need.
  - D Any temporary change during life.
7. Which statement about mutation is correct? [1]
  - A A directed change caused by need.
  - B Any temporary change during life.
  - C Mutation is a random change in genetic material.
  - D Organisms choosing useful mutations.

8. Which statement about adaptation and selection is correct? [1]
- A Any temporary change during life.
  - B Organisms choosing useful mutations.
  - C Random mating without selection.
  - D Adaptive features improve survival and reproduction in a particular environment.
9. Which statement about selective breeding is correct? [1]
- A Humans choose parents with desired phenotypes, breed them and select offspring over many generations.
  - B Organisms choosing useful mutations.
  - C Random mating without selection.
  - D Differences only between species.
10. Which prediction is most appropriate when a drought persists for many generations? [1]
- A Differences only between species.
  - B Any temporary change during life.
  - C Alleles for water-conserving features may increase.
  - D Random mating without selection.

## Section B · Structured questions

11 Biologists use precise vocabulary when studying variation and selection.

(a) Define variation.

[2]

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(b) Define mutation.

[2]

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(c) Distinguish adaptation from the incorrect idea: “any temporary change during life”.

[2]

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(d) State two important features of variation.

[2]

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**[Total: 8]**

12 A drought persists for many generations.

(a) State the most likely biological outcome.

[2]

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(b) Explain the outcome using a linked biological mechanism.

[3]

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(c) Predict what would happen if the relevant change were reversed. [2]

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(d) State one assumption made in this prediction. [1]

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**[Total: 8]**

13 A student records percentage survival under three conditions.

| condition          | percentage survival |
|--------------------|---------------------|
| susceptible strain | 2                   |
| mixed population   | 36                  |
| resistant strain   | 91                  |

(a) Calculate the difference between the highest and lowest recorded values. [2]

(b) Describe the pattern in the results. [2]

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(c) Explain the pattern using biological knowledge. [3]

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(d) State one limitation of drawing a conclusion from these data. [1]

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**[Total: 8]**

14 A student plans to measure variation in leaf length in one plant population.

(a) Identify the independent variable. [1]

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(b) Identify the dependent variable. [1]

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(c) State two control variables. [2]

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(d) Describe how to obtain reliable results across a suitable range. [3]

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(e) Suggest one specific improvement. [1]

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**[Total: 8]**

15 Apply your knowledge of variation and selection to an unfamiliar context.

(a) Define natural selection. [2]

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(b) Explain the biological importance of adaptation and selection. [3]

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(c) Correct this statement: “random mating without selection”.

[2]

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(d) State one further question that could be investigated.

[1]

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**[Total: 8]**

# Worked solutions

**Remove or fold back this section while completing the practice paper.**

## Marking routine

Mark each part separately. Award points for accurate biology, use of evidence and linked reasoning, then diagnose the first error.

## Solutions · Multiple choice

- 1 A. Differences between individuals of the same species.
- 2 B. A random change in genetic material.
- 3 C. An inherited feature increasing fitness in an environment.
- 4 D. Differential survival and reproduction caused by environmental pressures.
- 5 A. Human selection of parents with desired features.
- 6 B. Choose an appropriate graph: histogram for continuous data, bar chart for categories.
- 7 C. Mutations arise randomly; the environment selects among resulting phenotypes.
- 8 D. Use “individuals are selected; populations evolve”.
- 9 A. Evaluate both agricultural benefit and genetic risk.
- 10 C. Individuals with advantageous inherited variation survive and reproduce more.

## Solutions · Structured questions

11. (a) differences between individuals of the same species [2]. (b) a random change in genetic material [2]. (c) an inherited feature increasing fitness in an environment; not any temporary change during life [2]. (d) Credit any two accurate features from the guide: Continuous variation shows a range and is often controlled by many genes plus environment, such as height. Discontinuous variation has distinct categories and is often controlled by one or few genes, such as ABO blood group. Genetic variation arises from mutation, meiosis and fertilisation. [2].

12. (a) Alleles for water-conserving features may increase [2]. (b) Individuals with advantageous inherited variation survive and reproduce more [3]. (c) Accept the biologically justified opposite trend when the causal factor is reversed [2]. (d) One valid assumption, such as other relevant conditions remaining constant [1].

13. (a)  $91 - 2 = 89$  [2]. (b) The values change from 2 to 91, with the highest value 91. [2]. (c) Resistance alleles allow carriers to survive antibiotic selection and reproduce [3]. (d) Only three conditions / no repeats / small dataset / uncontrolled variation [1].

14. (a) Plant sampled [1]. (b) Leaf length [1]. (c) Same leaf position, maturity, measuring method and habitat area [2]. (d) Use several values across a sensible range, repeat each condition, identify anomalies and calculate means [3]. (e) Use random sampling and a larger sample size [1].

15. (a) differential survival and reproduction caused by environmental pressures [2]. (b) Award linked points from: Adaptive features improve survival and reproduction in a particular environment. Natural selection requires variation, selection pressure, differential survival and reproduction, inheritance and change in allele frequency over generations. Evolution is population change, not purposeful change by individuals. [3]. (c) Correct statement: selective breeding means human selection of parents with desired features [2]. (d) Any testable relevant biological question [1].

### Self-assessment

Classify the first lost mark as definition, biological knowledge, causal link, data use, experimental design or communication.

Evidence used: Cambridge IGCSE Co-ordinated Sciences 0654 syllabus 2025-2027 and June 2026 papers, mark schemes and Principal Examiner Report supplied by the user. All practice questions are newly written.